AI FOR EVERYONE — MINI-PROJECT

AI-Powered College FAQ Assistant

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Project Overview

OBJECTIVE & PROBLEM

Develop a conceptual AI assistant to help students quickly find answers to routine college FAQs, reducing the time spent searching multiple sources.

AI TOOLS INVOLVED

- **ChatGPT:** FAQ generation & refinement
- **Google Gemini:** Alternative response generation
- **Microsoft Copilot:** Information structuring

PROPOSED AI SOLUTION

An AI-assisted FAQ system that organizes common queries, generates clear answers, and directs students to official sources for verification.

EXPECTED OUTCOME

- Structured FAQ database for student support
- Demonstration of prompt engineering principles
- Evaluation of AI consistency in academic contexts

Problem Identification

Students face significant challenges when searching for routine college information, leading to wasted time and confusion.

Information Fragmentation

Crucial details are scattered across physical notice boards, official websites, emails, and messaging groups, requiring excessive search time.

Outdated Content

Risk of students relying on obsolete examination schedules or fee structures due to a lack of a centralized, easily accessible knowledge base.

Repetitive Inquiries

College administration often handles hundreds of identical routine queries, which could be efficiently automated through an AI FAQ system.

Proposed Solution & Workflow



The workflow ensures that while AI provides **speed** and **clarity**, human oversight maintains **accuracy**. All AI-generated responses are derived from verified input data and must be cross-checked against official college sources before being considered final.

Objectives & Scope

Project Objectives

Efficiency
Provide instant responses to routine student queries, minimizing search effort.

Consistency
Ensure uniform information delivery across various student departments.

Organization
Transform fragmented documents into a structured, searchable FAQ dataset.

Prototype Boundaries

Conceptual Stage
This project is a functional prototype demonstration, not a deployed application.

Factual Limitations
Does not substitute official college notices or personalized administrative advice.

Future Scope
Potential for web integration as a verified chatbot with live backend data.

AI Tools & Their Roles

AI TOOL	PRIMARY TASK	INPUT PROVIDED	CONTRIBUTION & CONTRIBUTION
ChatGPT	Problem analysis & FAQ generation	College policy summaries & student personas	Drafted comprehensive Q&A sets and refined the assistant's personality. *Requires verification of specific dates/fees.*
Google Gemini	Response comparison & simplification	ChatGPT-generated outputs	Evaluated clarity and provided alternative, student-friendly phrasing. *May generalize complex administrative rules.*
MS Copilot	Data structuring & table creation	Raw Q&A text strings	Organized unstructured data into clean, manageable HTML/Excel-ready tables. *Formatting may require manual adjustment.*

Prompt Engineering & Input Design

Key Principles

Persona Adoption

Act as a professional student-support assistant to ensure a helpful and formal tone.

Contextual Guardrails

Explicitly forbid the AI from inventing facts or institutional data not provided in the source.

Output Structuring

Request specific formats (e.g., Markdown tables) to ensure data usability.

SAMPLE PROMPT 1: EXTRACTION

```
"Extract all information related to examination schedules and attendance rules
from the provided text. Present it as a clear Q&A list for new students."
```

Focuses the AI on relevant academic data while filtering noise.

SAMPLE PROMPT 2: REFINEMENT

```
"Convert the following table into student-friendly language. If any specific
date or amount is missing, use placeholders like [Check Official Notice]."
```

Ensures factual safety and improves accessibility for students.

Illustrative Output: FAQ Dataset

CATEGORY	STUDENT QUESTION	AI-GENERATED RESPONSE (CONCEPTUAL)
Academics	What is the minimum attendance required?	A minimum of 75% attendance is mandatory. **[Refer to official department notice for relaxation rules]**.
Examinations	When will the mid-semester exams start?	Exams typically start in the second month of the semester. **[Check the current semester's academic calendar]**.
Hostel	How do I apply for hostel accommodation?	Submit the hostel application form via the online portal. **[Visit the hostel warden's office for seat availability]**.
Fees	Where can I pay the tuition fees?	Fees can be paid online through the college portal or via bank challan. **[Verify the latest fee amount from the accounts office]**.
Admission	What documents are needed for verification?	Original certificates, identity proof, and allotment letter. **[Refer to the official admission brochure for the full list]**.

Illustrative Tool Comparison

Sample Question: "What happens if I miss my exam due to a medical emergency?"

ChatGPT-4o (Draft)

> If you miss an exam due to illness, you must submit a medical certificate from a registered practitioner to the Examination Cell within 3 days. A re-exam may be scheduled based on the Principal's approval.

Observation: Detailed and structured, but assumes specific local procedures without verification.

Google Gemini (Simplification)

> In case of medical emergencies, please inform your department immediately. Submit your medical reports to the office. [Refer to Official College Rulebook for exact submission deadlines].

Observation: Safer response with explicit source placeholders, though slightly less detailed on specific steps.

Conceptual Prototype Demonstration

College FAQ Assistant

How can I apply for an internship NOC?

To apply for an internship NOC (No Objection Certificate):
1. Obtain a letter from your internship company.
2. Submit the letter and a formal application to the HOD.
3. Once approved, the office will issue the NOC.

Verified against Academic Handbook 2024

1. INPUT CONTEXT

The AI is supplied with the "Administrative Procedures" section of the student handbook.

2. RESPONSE LOGIC

AI extracts the multi-step process and re-formats it into a readable list for the student.

3. VERIFICATION

The "Verified" tag is only added when the output perfectly matches the source document.

Testing Plan & Evaluation

TEST QUESTION	EXPECTED INFORMATION	AI RESPONSE STATUS	FACTUAL CORRECTNESS	ACTION REQUIRED
Fee payment deadline?	*Date from official notice*	*[Pending]*	*[Pending]*	*Verify against receipt.*
Hostel leave procedure?	*Gate pass rules*	*[Pending]*	*[Pending]*	*Check warden's rules.*
Backlog exam form link?	*Official portal URL*	*[Pending]*	*[Pending]*	*Test URL validity.*
Scholarship eligibility?	*Criteria list*	*[Pending]*	*[Pending]*	*Cross-check with govt.*
Library timings?	*Opening/Closing hours*	*[Pending]*	*[Pending]*	*Confirm with librarian.*

Key Takeaways, Safety & References

LIMITATIONS & SAFETY

- Risk of AI hallucinations in specific dates.
- Human verification is mandatory.
- No sensitive student data was used in prompts.

ACADEMIC REFERENCES

1. AI Tools Documentation:
OpenAI (2024). ChatGPT-4o System Card. openai.com
Google (2024). Gemini Pro Technical Report. deepmind.google
2. Institutional Sources:
Institutional website: not supplied
Academic handbook: not supplied
3. Methodology:
Prompt examples are original instructional content for this mini-project.

Thank You!